

Sai Charan Merugu

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SKILLS & INTERESTS:

Agentic Systems & Generative AI: Multi-Agent Orchestration, Agent Workflows, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Prompt Engineering, Function Calling, Structured Output Control, Autonomous Reasoning Pipelines

Model Development & Optimization: Model Fine-Tuning, LoRA / PEFT, Quantized Models, Multimodal Models, Transformer Optimization, Low-Latency Inference

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Computer Vision & Speech Systems: Vision Models, Image Processing, Object Detection, OCR Pipelines, Low-Latency Text-to-Speech (TTS), Speech-to-Text (STT), Real-Time Speech Pipelines

Retrieval & Search Systems: FAISS, Vector Databases, Semantic Search, Embedding Pipelines, Similarity Search

Frameworks & Applied AI Systems: LangChain, LangGraph, AutoGen, Hugging Face Transformers, OpenAI SDK, Unreal Engine Integration, Simulation-Based AI Systems

Backend, Cloud & Infrastructure: Python, FastAPI, REST APIs, SQL, Docker, AWS, Azure, GCP, CI/CD, Git, Linux

Data Engineering: Data Preprocessing, ETL Pipelines, Large-Scale Dataset Handling, Data Wrangling, Data Cleaning

PROFESSIONAL EXPERIENCE

Handshake AI

California, US (remote)

Applied AI Research Engineer

Feb 2026 – Present

- Designed and refined domain-specific prompts for Large Language Models (LLMs) to improve response quality, reasoning consistency, and task alignment across research workflows.
- Evaluated LLM-generated outputs for accuracy, relevance, factual consistency, and adherence to domain-specific standards.
- Performed iterative prompt optimization through structured testing, response comparison, and failure analysis to improve model behavior.
- Contributed to milestone-based research deliverables in a remote, asynchronous environment, maintaining quality and turnaround expectations.
- Documented prompt patterns, evaluation observations, and output quality trends to support reproducible research outcomes.

Future State University

California, US (remote)

Associate AI Engineer

Nov 2025 – Feb 2026

- Built custom RAG systems integrating transformer models with structured business datasets.
- Developed AI agents with contextual memory for workflow automation across departments.
- Deployed production-grade AI models with monitoring, validation, and performance tracking.
- Integrated AI solutions into internal systems to optimize operational processes.
- Designed experimentation pipelines ensuring reproducibility and business-aligned evaluation metrics.

Arizona State University

Tempe, Arizona, US (remote)

AI Researcher

March 2024 – Sept 2025

- Built Unreal Engine-integrated AI systems with low-latency TTS and STT pipelines, RAG-based retrieval, domain-specific model fine-tuning, and FAISS-powered fast indexing to support real-time multimodal AI workflows.
- Developed RAG-based models for context-aware retrieval and response generation over structured and unstructured datasets.
- Performed model fine-tuning for domain-specific performance improvement, optimizing accuracy, latency, and inference quality.
- Implemented fast vector indexing with FAISS to accelerate semantic search and retrieval in large-scale knowledge systems.
- Worked on multimodal and quantized model optimization to improve inference efficiency and support production-scale deployment.
- Engineered low-latency model serving infrastructure with optimized batching and hardware-aware scheduling.

Infosys LTD

Hyderabad, India

Machine Learning Engineer

August 2022 – August 2023

- Developed machine learning microservices using Python, TensorFlow, and PyTorch to support enterprise automation workflows.
- Built ML models for structured and semi-structured business data to automate classification, tagging, and workflow routing.
- Deployed containerized ML services across cloud environments using Docker on Amazon Web Services and Microsoft Azure.
- Implemented monitoring pipelines for model drift detection, performance validation, and reliability tracking in production systems.
- Worked with engineering and business teams to integrate AI-driven automation solutions into enterprise applications.

Darwin Box

Hyderabad, India

Data Scientist

March 2022 – Aug 2022

- Built machine learning models (regression and classification) for predictive analytics and business optimization use cases.
- Performed data cleaning, feature engineering, and preprocessing for structured and semi-structured datasets.
- Evaluated model performance using cross-validation, ROC-AUC, precision-recall, and statistical testing methods.
- Automated reporting and analytics workflows to improve operational efficiency.

Education

University of north texas(graduated May 2025): Denton,Texas,USA

Master's in computer and information science

Sreenidhi institute of Science and Technology (July 2018-July 22) Hyderrabad,INDIA

Bachelor's in computer science